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Corrosion characteristics of high entropy alloys (HEAs)

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The present paper reports on the electrochemical properties of a wide range of high entropy alloys (HEAs) in a 0·6 M NaCl solution. A consolidated treatise of the topic has to date been lacking, and the purpose of the work herein is to present a primitive galvanic series for numerous HEAs, along with a broad survey of results typifying their electrochemical characteristics, passivity and comparative electrochemistry. The results are coupled with microstructural characterisation. The range of potentials for HEAs is comparable to or nobler than austenitic stainless steel, with a number of HEAs displaying higher pitting potential (E_{pit}) values than stainless steels, in spite of possessing heterogeneous microstructures.

Keywords: High entropy alloy, Corrosion, Galvanic series, Polarisation, Electrochemistry, Microstructure

Introduction

Structural alloys including steels, aluminium, magnesium, titanium or nickel based alloys, are typically based on alloy design centred on controlling the phase evolution from one or two principle alloying elements. While additional elements are often added to many alloy systems for the purposes of inoculation, grain refinement, corrosion resistance, scavenging or melt protection, such additions are not intended to interfere with the phase evolution.[1–2] A decade ago, Yeh *et al.*[3] advanced the concept of so called high entropy alloys (HEAs), which are alloys composed of five or more principle elements in equimolar or near equimolar ratios with concentration of each element in the range of 5–35 at-%.

The unique aspect of HEAs that has aroused interest in such systems is that, instead of forming ordered intermetallic phases, they seem to have the tendency to form solid solutions, considered to be stabilised by the high entropy of mixing.[3] However, pursuit of HEAs has indicated that the high entropy of mixing is in itself not enough to assure the formation of a random solid solution,[4–5] with only selected compositions (in the range of what is considered a HEA) forming what can be termed as simple solid solutions. Zhang *et al.*[6] proposed that in order to form solid solutions, high entropy alloys should be composed of elements that possess small differences in atomic size, along with the absolute enthalpy of mixing being near zero. Reports suggest atomic size differences $<6\cdot5\%$, an enthalpy of mixing in the range of -15 to $5\ \mathrm{kJ\ mol^{-1}}$, while the entropy of mixing should be in the range of ~12–$17\cdot5\ \mathrm{J\ K^{-1}\ mol^{-1}}$. HEAs have been reported with high hardness and high compressive strength, both at room temperature and at elevated temperature.[3,7–13] They have also been reported to possess outstanding resistance to wear and oxidation[13–14] and high thermal stability,[15] which has stimulated significant interest in HEAs as promising candidates for many applications.

In terms of the corrosion performance of HEAs, aqueous corrosion testing has also been reported over the past decade.[13,16–23,25–37] In one of the earliest studies since the discovery of HEAs, Chen *et al.*[16] compared the electrochemical properties of a $Cu_{0\cdot5}NiAlCoCrFeSi$ HEA with 304 stainless steel (AISI type 304 SS), reporting that over a wide range of NaCl and H_2SO_4 (sulphuric acid) concentration, the resistance to general corrosion of $Cu_{0\cdot5}NiAlCoCrFeSi$ was superior to that of 304SS, irrespective of the electrolyte concentration varying from 0·1 to 1 M. However, the resistance to pitting corrosion of $Cu_{0\cdot5}NiAlCoCrFeSi$ was reported to be inferior to that of 304SS in a chloride environment on the basis of anodic polarisation testing and SEM images of the HEA after an anodic polarisation.[16] In the case of as cast AlCoCrFeNi, it has been reported that, due to the segregation of Cr to interdendritic regions and also to fine precipitates within the dendrites, the Cr depleted dendrites suffer from selective corrosion in 0·6 M NaCl.[17] This phenomenon would be akin to sensitisation in stainless steels where localised corrosion occurs at sites of Cr depletion; however, such phenomena are significantly less studied in HEAs and have only been reported in this specific case.[17] In a similar alloy with Cu additions, Qiu[18] reported that AlCrFeNiCoCu demonstrated excellent corrosion resistance in 1 M NaCl, compared to 304SS, with the AlCrFeNiCoCu HEA presenting a more noble potential and a corrosion current density about an order of magnitude lower than 304SS. In the Al free variant of this alloy, Hsu *et al.*[19] investigated the corrosion behaviour of the $FeCoNiCrCu_x$ system and noted that an increased proportion of Cu deteriorated the corrosion resistance of $FeCoNiCrCu_x$ which they attributed to microgalvanic

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corrosion caused by segregation of Cu. The suggestion was that Cu rich interdendritic regions were less noble than the Cu depleted dendrites; therefore, the Cu rich regions were dissolved preferentially.

For the $Al_{0·5}CoCrCuFeNiB_x$ system studied in 1N H_2SO_4, the general corrosion resistance of $Al_{0·5}$CoCrCuFeNi was found to be superior to that of 304SS.[20] It was also determined that the B lean HEAs performed best in terms of corrosion resistance, as the addition of B led to decreased corrosion resistance purported to be due to the formation of Cr borides. However, $Al_{0·5}CoCrCuFeNiB_x$ was noted as not being prone to localised corrosion.[20]

In comparing within a given class of HEAs, it was reported that an increased proportion of Al in $Al_xCrFe_{1·5}MnNi_{0·5}$ alloys led to a higher general corrosion susceptibility of HEAs in a 0·5 M H_2SO_4 solution. In a 1 M NaCl solution, the pitting potentials of $Al_{0·5}CrFe_{1·5}MnNi_{0·5}$ HEAs are significantly lower than Al free $CrFe_{1·5}MnNi_{0·5}$ by ~ 140 mV.[21] Kao *et al.*[22] also examined the effect of Al additions on the corrosion of HEAs, indicating that a higher amount of Al in Al_xCoCrFeNi alloys was detrimental to corrosion resistance in 0·5 M H_2SO_4 at temperatures higher than 27°C. Chou *et al.*[23] examined the influence of molybdenum on the corrosion resistance of $Co_{1·5}CrFeNi_{1·5}Ti_{0·5}Mo_x$ alloys in 0·5 M H_2SO_4 and 1 M NaOH solutions. Interestingly, in contrast to conventional stainless steels, the authors suggested that the corrosion resistance of the Mo free HEA was superior to the Mo containing alloys. However, in the case of a 1 M NaCl solution, the Mo containing HEAs were less susceptible to pitting corrosion compared with the Mo free alloy as judged by cyclic potentiodynamic polarisation testing.[23] The effect of Ni additions on the corrosion of the $Al_2CrFeCoCuTiNi_x$ system has also been investigated. In both 0·6 M NaCl and 1 M NaOH, the corrosion of $Al_2CrFeCoCuTiNi_x$ alloys decreased with Ni addition ($x \leqslant 1$) and then increased with increasing Ni ($x \geqslant 1$),[13] a response that is again counter to the wisdom in the behaviour of stainless steels.[24] The addition of Nb to CoCrCuFeNi was reported to promote a lower corrosion current density.[25] In the case of the many element $Al_2CrFeCoCuNiTi_x$ system,[26] HEA variants containing Ti demonstrated lower corrosion rates than Ti free HEA in a 0·5 M HNO_3 solution. Conversely, for $Al_{0·3}CrFe_{1·5}MnNi_{0·5}Ti_x$ and $Al_{0·3}CrFe_{1·5}MnNi_{0·5}Si_x$ systems, the addition of small amounts of Ti ($x = 0·2$) or Si ($x = 0·2$) deteriorated the general corrosion resistance of the HEAs.[27]

Based on the abridged review above, it is noted that while several corrosion related studies on HEAs exist, such as those in the limited list herein, the complexity of the alloy systems makes a first order analysis of general trends across different HEA systems very difficult. For example, in some cases, Al is found to be beneficial, in others detrimental. This is determined by the particular constituents of the remaining composition. This is similarly the case for Cu, Ti and Si. Furthermore, a wider comparison of HEA corrosion across systems is yet to be reported; moreover, the correlation between processing, thermal history and microstructure, individually and in combination, is all in their infancy. Aspects of electrochemistry also need to be understood on the basis that HEAs have the potential for significant segregation of solute and the existence of several phases (of widely varying composition) simultaneously, and do not follow the conventional golden rules of stainless steel with regards to the influence of Cr, Ni or Mo.

This paper conducts an initial consolidated summary leading to a primitive galvanic series for select HEAs of widely varying composition in 0·6 M NaCl. This is accompanied by a discussion of the corrosion potentials, passivity and electrochemical characteristics with reference to the polarisation response and alloy microstructure of a range of HEA systems.

Experimental

Material preparation

The HEAs reported herein were produced by a number of routes, each unique on the basis of capability of consolidating metals with wide melting point differences. As such, conventional casting was not employed.

HEAs were produced by laser melting (LM). In such cases, metal powders (with purity higher than 99·5 wt-% and typical size of 50–100 µm) were used. The LM of powders was carried out using a Trumpf TruLaser Cell 7040, with melting performed under an argon atmosphere.

Production of HEAs using mechanical alloying was also carried out. In such cases, metal powders were blended in the appropriate proportion using high energy ball milling. The milled powders were then consolidated fully using spark plasma sintering (SPS) (DR.SINTER model SPS-925).

Additionally, arc melting (AM) was also employed to produce HEAs. This technique employed conventional AM in a reduced argon atmosphere. The HEAs produced were melted on a water cooled copper hearth, with the starting material being pure (99·99 wt-%) metal chunks.

Electrochemical tests

Potentiodynamic polarisation testing was conducted in a typical three-electrode flat cell (Princeton Applied Research) with the specimen as the working electrode, a saturated calomel reference electrode (SCE) and a platinum mesh as a counter electrode. The test solution was in each case quiescent 0·6 M NaCl, and testing was at 25°C. Before testing, the specimens were ground to 2000 grit finish (SiC paper), and at least four replicate tests were conducted for each alloy. The open circuit potential (OCP) was measured for 30 min before polarisation testing to achieve an approximately stable potential. Each specimen was scanned potentiodynamically at a rate of 1 mV s^{-1} from the initial potential of -150 mV_{SCE} versus OCP to the final potential of 2·0 V_{SCE} using a Biologic VMP potentiostat under the control of EC-Lab 10·02 software. The corrosion potentials (E_{corr}) and corrosion current densities (i_{corr}) were estimated by user based analysis via EC-lab.

Microstructural analysis

The constituent phases of the HEAs were determined using X-ray diffraction (XRD). This was done using an X-ray diffractometer (Phillips PW1130), with a Cu target, a voltage of 40 kV, current of 25 mA and a scan range from 10° to 100° with a speed of 1·5° min^{-1}. Known compounds in the powder diffraction database were not employed to identify phases for HEAs, as the phases (and their unique composition) in HEAs are largely unknown. Additionally, the interplanar spacing (d-spacing) of HEAs are unique and not in the powder diffraction database. Instead, we calculate the d-spacing for individual peaks in the collected XRD pattern and match the d-spacing according to the selection rules for the relevant crystal structure. For example, for an fcc structure, the d-spacing of peaks in XRD pattern shall refer to the (111), (200), (220) and (311) plane.

As such, what we are most concerned with (namely for homogenous HEAs) is the crystal structure (i.e. fcc or bcc). The following is how we determine the crystal structure of phases. We calculate the dsp for individual peaks in XRD pattern and see if these dsp matches the selection rules for fcc or bcc structure. For example, for fcc structure, dsp of peaks in XRD pattern should refer to (111), (200), (220) and (311) plane.

Specimens were also investigated via scanning (SEM) and transmission electron microscopy (TEM). SEM samples were polished to a 1 μm diamond paste finish and then imaged using a FEI Nova NanoSEM 450 in the backscattered electron (BSE) mode. TEM specimens were prepared by focused ion beam milling, using a Quanta 3D FIB. A thin foil was milled from a bulk HEA specimen via Ga ion milling, followed by 'lift out' onto a Cu grid, and followed subsequently by a thinning process of the foil to ∼100 nm in thickness. TEM imaging was carried out using an FEI Tecnai G2 T20 TEM.

Results and discussion

Galvanic series and pitting potential of high entropy alloys

Based on results from potentiodynamic polarisation testing, the summary of the corrosion potentials (E_{corr}) and pitting potentials (E_{pit}) for a selection of HEAs in a quiescent 0·6 M NaCl solution at 25°C (a standardised electrolyte to allow comparison with published galvanic series) is shown in Fig. 1.

The E_{corr} of these HEAs is in a range of ~ -498 to -180 mV$_{SCE}$. As anticipated, the HEAs presented corrosion potentials that are significantly nobler than those of mild steel and aluminium alloys. The corrosion potentials of some of the select HEAs were in a band comparable to that between ferritic and austenitic stainless, with some HEAs displaying even more noble potentials than austenitic stainless steels. The E_{pit} of most HEAs studied herein is more positive than that of ferritic stainless steels. On the basis of discrimination of pitting susceptibility from E_{pit} alone, several HEAs demonstrate a higher (more noble) E_{pit} than austenitic stainless steel, particularly $Ti_{0·3}(CoCrFeNi)_{0·7}$ alloy, which displays the very noble E_{pit} of 1040 mV$_{SCE}$.

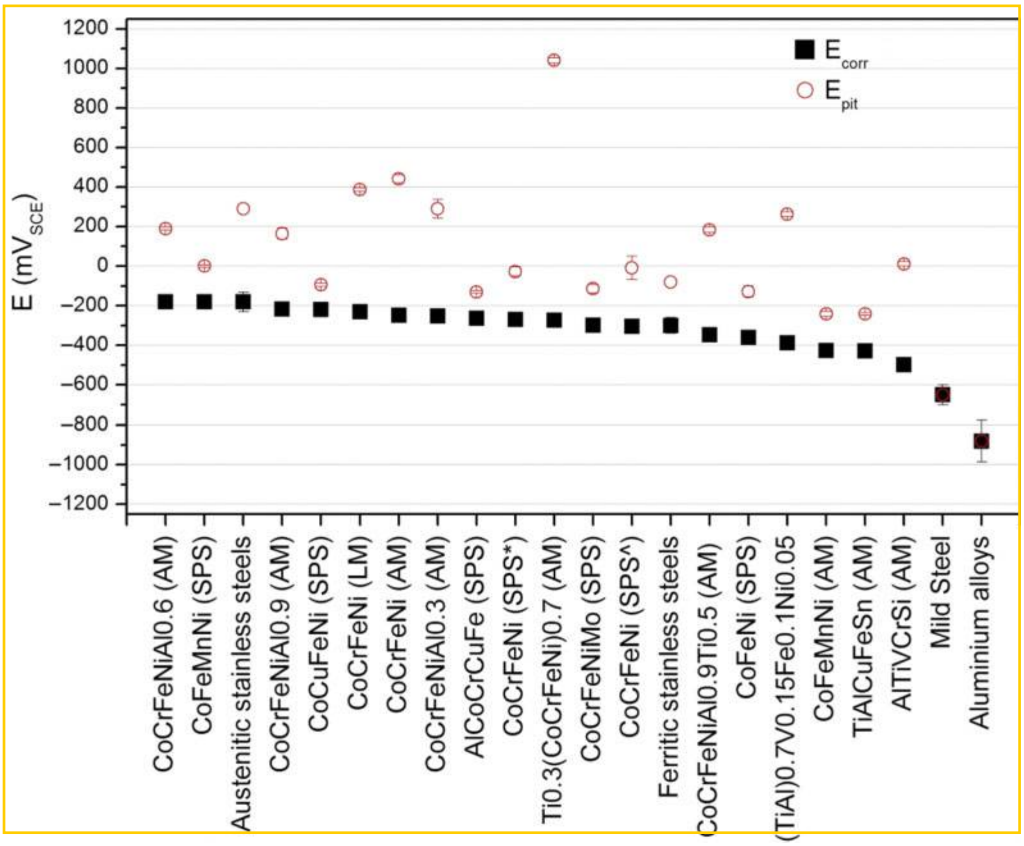
Some general observations from Fig. 1 include that corrosion potentials of the HEAs are not necessarily in the range expected from their constituent metals. For example, for the HEA AlCoCrCuFe, one would expect that the E_{corr} would be below $-0·35$ to $-0·4$ V$_{SCE}$ (corresponding to the known OCPs of Cr, and Cu in chloride solutions, being the most noble elements in this HEA combination). This shows that the HEA forms heterogeneous and unique phases that have their own electrochemical characteristics, independent of their constituent metals. If one assumes CoFeNi (SPS) as the baseline for analysis, it can be seen that the addition of Cr ennobles the HEA (more noble E_{corr}) and improves the HEA resistance to pitting (higher E_{pit}). This can be inferred by comparing the E_{corr} and E_{pit} of the HEA with the composition CoCrFeNi (Fig. 1). The incorporation of Ti into CoCrFeNi further increases the E_{pit} as observed for the alloy $Ti_{0·3}(CoCrFeNi)_{0·7}$. The addition of Al to the HEA with the base CoCrFeNi makes them more noble, as inferred from the corrosion potentials of $Al_{0·3}CoCrFeNi$ (AM), $Al_{0·6}CoCrFeNi$ (AM) or $Al_{0·9}CoCrFeNi$ (AM), albeit that Al is the least noble element in all the HEAs studied (pure Al having its own characteristic potential of ~ -1 V$_{SCE}$ (in 0·6 M NaCl).

It is also noteworthy that there is a significant difference in pitting potentials between HEAs made by SPS and HEAs with the identical composition made by AM. CoCrFeNi (AM) exhibits an E_{pit} of 442 mV$_{SCE}$, while CoCrFeNi (SPS*) and CoCrFeNi (SPS^) have a much lower E_{pit} (-28 and -8 mV$_{SCE}$ respectively). The possible reasons for these characteristics are discussed in the following sections.

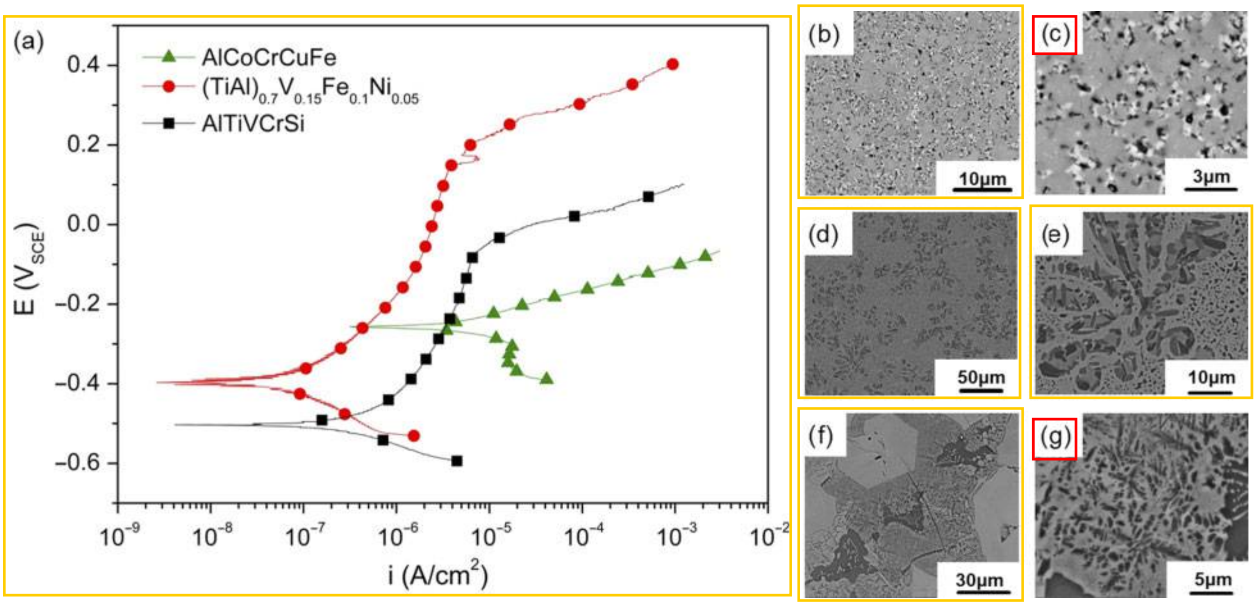
Electrochemical response and characterisation of select high entropy alloys

In order to consolidate the survey of the electrochemical properties of the HEAs reported herein, the alloys have been separated on the basis of some key characteristics. In the case of the data shown in Fig. 2, these three HEAs were selected on the basis that they present the most heterogeneous microstructures among the reported HEAs (their associated microstructures are seen in Fig. 2*b*–*g*). Such microstructures are dramatically different to stainless steels and significantly more heterogeneous than most conventional structural alloys.

The potentiodynamic polarisation curves of as cast AlCoCrCuFe, $(TiAl)_{0·7}V_{0·15}Fe_{0·1}Ni_{0·05}$ and AlTiVCrSi in a 0·6 M NaCl solution are shown in Fig. 2*a*. The tabulated values of extracted parameters compared to 304L (from Hsu *et al.*[19]) are listed in Table 1, which include corrosion potential E_{corr}, corrosion current density i_{corr}, pitting potential E_{pit} and the potential range of the passive region ($E_{corr} - E_{pit}$). The comparison to 304L was made on the basis that the HEAs are Mo free and C free. The electrochemical data reveal that these alloys are 'spontaneously passive'. This is particularly obvious for $(TiAl)_{0·7}V_{0·15}Fe_{0·1}Ni_{0·05}$ and AlTiVCrSi, but less so for AlCoCrCuFe. The latter alloy has significantly more rapid cathodic kinetics, and hence a significantly higher E_{corr}. The enhanced cathodic kinetics of AlCoCrCuFe also results in a higher corrosion current density (i_{corr}). The window of passivity on AlCoCrCuFe (from repeat tests) is as much as ~ 130 mV; however it is between 600 and 650 mV$_{SCE}$ for both AlTiVCrSi and $(TiAl)_{0·7}V_{0·15}Fe_{0·1}Ni_{0·05}$. The latter alloys reveal a clear and wide window of passivity, in spite of the 0·6 M NaCl electrolyte and a heterogeneous microstructure, which is a key point. From an electrochemical perspective, such passivity in the presence of a multiphase microstructure is somewhat unprecedented in conventional alloys or stainless steels (perhaps with the exception of some superalloys or Zr alloys). The smaller window of passivity in the case of AlCoCrCuFe is perhaps associated with the presence of Co, and the intense segregation of heavier elements to the regions of light contrast as seen in Fig. 2*c*. Previous studies on Cu containing HEAs reveal that copper has the



1 Summary of HEA corrosion potential (E_{corr}) in mV$_{SCE}$ and pitting potential (E_{pit}) in mV$_{SCE}$ in quiescent 0·6 M NaCl at 25°C; CoCrFeNi (SPS*) was held at 1000°C for 5 min, while CoCrFeNi (SPS^) was firstly held at 500°C for 5 min and then sintered at 1000°C for another 5 min; E_{corr} and E_{pit} data of steels and aluminium alloys are obtained from previous studies[24,38–39]



2 *a* polarisation curves for as cast AlCoCrCuFe, $(TiAl)_{0·7}V_{0·15}Fe_{0·1}Ni_{0·05}$ and AlTiVCrSi in 0·6 M NaCl at 25°C; *b* BSE image of AlCoCrCuFe at low magnification; *c* BSE image of AlCoCrCuFe at higher magnification; *d* BSE image of $(TiAl)_{0·7}V_{0·15}Fe_{0·1}Ni_{0·05}$ at low magnification; *e* BSE image of $(TiAl)_{0·7}V_{0·15}Fe_{0·1}Ni_{0·05}$ at higher magnification; *f* BSE image of AlTiVCrSi at low magnification; *g* BSE image of AlTiVCrSi at higher magnification

tendency to segregate into the interdendritic regions to form a Cu rich fcc phase between Cu depleted dendrites.[9,19,40–41] Table 2 lists the binary mixing of enthalpy (ΔH_{mix}) of Cu with other elements. These values of ΔH_{mix} are either very close to zero or fairly positive. This indicates that there is a weak bonding force between Cu and the other elements. Therefore, Cu tends to segregate into interdendritic regions. The XRD data for as cast AlCoCrCuFe are provided in Fig. 3. AlCoCrCuFe consists of a Cu rich fcc phase together with a bcc phase (Cu depleted matrix). When AlCoCrCuFe is exposed to an aqueous environment, the Cu depleted matrix is

Table 1 Electrochemical data associated with general corrosion and pitting corrosion of HEAs and 304L stainless steel in quiescent 0·6 M NaCl at 25°C

Sample	E_{corr}/mV$_{SCE}$	i_{corr}/μA cm^{-2}	E_{pit}/mV$_{SCE}$	Average $E_{pit}-E_{corr}$/mV
AlCoCrCuFe	−264±5	0·967±0·22	−130±5	135
$(TiAl)_{0\cdot7}V_{0\cdot15}Fe_{0\cdot1}Ni_{0\cdot05}$	−388±12	0·037±0·01	263±10	651
AlTiVCrSi	−498±20	0·168±0·03	11±10	510
304L[19]	−250	0·601	230	480

Table 2 Binary mixing of enthalpy of Cu with Co, Cr and Fe[42]

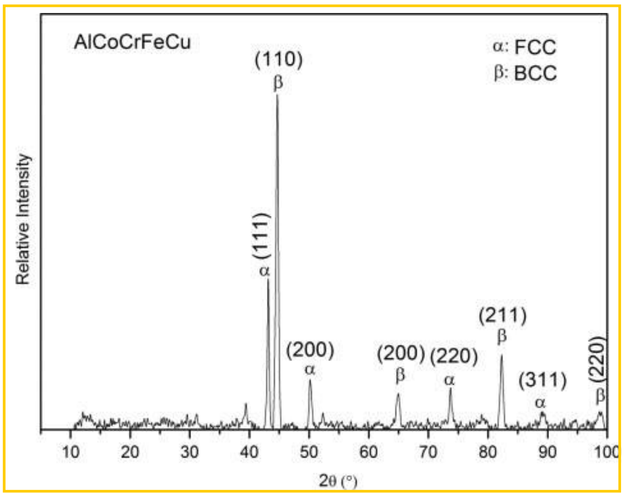
Element pair	ΔH_{mix}/kJ mol^{-1}
Cu–Al	−1
Cu–Co	6
Cu–Cr	12
Cu–Fe	13

comparatively nobler than the Cu rich interdendritic regions; therefore, the Cu rich phase may be a site for preferential dissolution, with such aspects discussed in the following section.

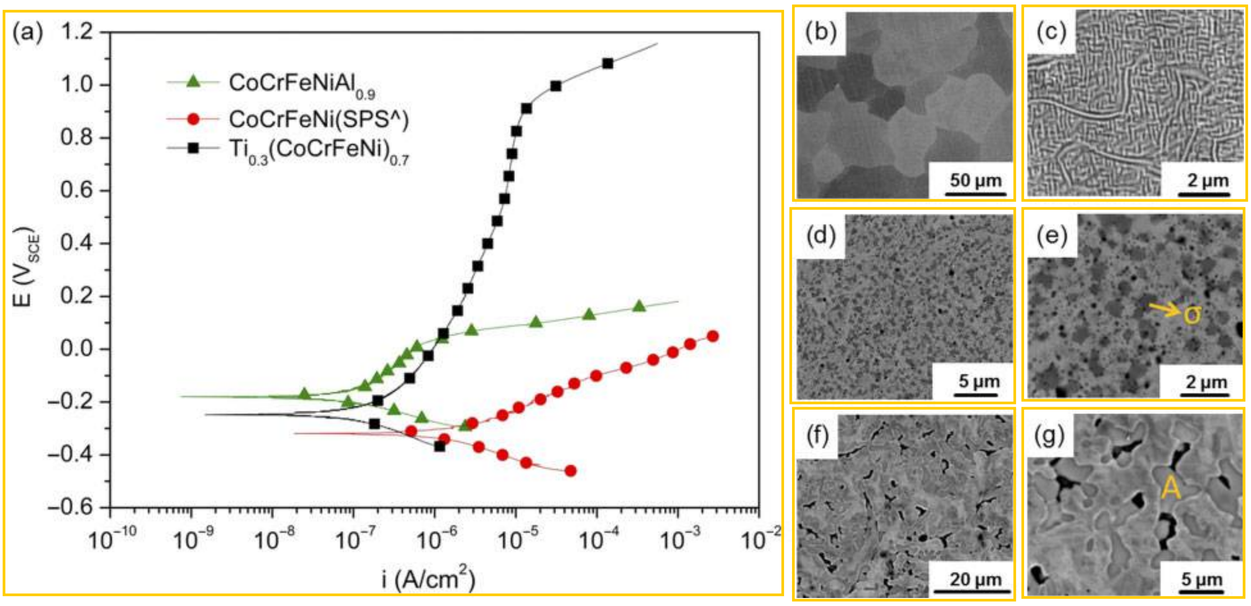
In considering a family of HEAs that are of relatively similar composition, but which present a significant difference in electrochemical performance (especially in terms of passivation), the polarisation response of such a selection of alloys is shown in Fig. 4. The alloys presented (Fig. 4) are $CoCrFeNiAl_{0\cdot9}$, CoCrFeNi (SPS^) and $Ti_{0\cdot3}(CoCrFeNi)_{0\cdot7}$ in 0·6 M NaCl. As with all the HEAs surveyed, the alloys shown in Fig. 4 are 'spontaneously passive' in 0·6 M NaCl. In addition, these alloys all have very similar cathodic reaction kinetics. The cathodic kinetics are more rapid than those of $(TiAl)_{0\cdot7}V_{0\cdot15}Fe_{0\cdot1}Ni_{0\cdot05}$ and AlTiVCrSi (Fig. 2), and hence, it is posited that the presence of Co enhances the cathodic kinetics and concomitantly raises the E_{corr}.

The microstructures of $CoCrFeNiAl_{0\cdot9}$, CoCrFeNi (SPS^) and $Ti_{0\cdot3}(CoCrFeNi)_{0\cdot7}$ are also provided in Fig. 4, while the associated electrochemical parameters are presented in Table 3. The difference in the electrochemical properties of the alloys is that the passive window $(E_{pit}-E_{corr})$ size scales according to $Ti_{0\cdot3}(CoCrFeNi)_{0\cdot7} > CoCrFeNiAl_{0\cdot9} > $ CoCrFeNi (SPS^). This result is interesting on the basis of the conventional wisdom that relates to corrosion and merits comment. The largest window of passivity correlates to the alloy variant that contains Ti, which is conventional in its own right. However, it is unique when considered with the microstructure. The very large passive window of $Ti_{0\cdot3}(CoCrFeNi)_{0\cdot7}$ persists in spite of a very heterogeneous microstructure (at least three unique phases) and significant segregation. The phase denoted as 'A' in the BSE image of $Ti_{0\cdot3}(CoCrFeNi)_{0\cdot7}$ (Fig. 4*g*) is enriched in Ti (∼62·8 at-%), with the associated energy dispersive spectroscopy (EDS) point analysis shown in Table 4. In the case of CoCrFeNi (SPS^), the microstructure is also multiphase, with the fine black dots in Fig. 4*e* that are actually synonymous with very fine pores. As noted, CoCrFeNi (SPS^) exhibits a significantly lower E_{pit} compared to CoCrFeNi (AM). In addition, the i_{corr} of CoCrFeNi (SPS^) (0·610 μA cm^{-2}) is significantly larger than that of CoCrFeNi (AM) (0·108 μA cm^{-2}). CoCrFeNi (AM) is a single fcc solid solution, as indicated by the XRD spectrum in Fig. 5*a*. On the other hand, CoCrFeNi (SPS^) exhibits a multiple phase microstructure with the corresponding XRD spectrum, as shown in Fig. 5*b*, indicating that it consists of an fcc and sigma type phase (σ). Praveen *et al.*[43] reported that CoCrFeNi sintered at 1000°C using SPS consisted of a major fcc phase and a minor Cr rich sigma phase. This sigma phase could be a CrFe/CrCo/CrFeCo intermetallic compound.[43] EDS mapping results for CoCrFeNi (SPS^) are provided in Fig. 6, confirming that the sigma phase is rich in Cr, while the fcc matrix is Cr depleted. In austenitic stainless steel, the σ phase is an intermetallic compound with a complex tetragonal structure. The σ phase is rich in Cr, with a typical normalised composition of the σ phase being Fe (∼59 wt-%), Cr (∼38 wt-%) and Ni (∼3 wt-%).[44] In stainless steel, Cr rich sigma phase has a detrimental effect on the corrosion resistance including pitting resistance.[24] Similar to the effect of Cr rich sigma phase in stainless steel, the evolution of a Cr rich σ phase in CoCrFeNi (SPS^) is also detrimental. The Cr rich σ phase is nobler than the adjacent Cr depleted fcc matrix, therefore rendering the Cr depleted matrix susceptible to corrosion.

In the case of $CoCrFeNiAl_{0\cdot9}$, this alloy exhibits a fine interconnected microstructure, which is unique on the basis of the connectivity of microstructural features. $CoCrFeNiAl_{0\cdot9}$ is comparatively very noble in 0·6 M NaCl and also exhibits a low corrosion current density compared to the other HEAs listed in Tables 1 and 3. In contrast to the posit that the presence of Co enhances cathodic kinetics and concomitantly raises the E_{corr}, a simpler microstructure is also associated with a high E_{corr}, albeit on the basis of reduced anodic kinetics. The instance of reduced anodic kinetics correlates with a low corrosion current density, as depicted by $CoCrFeNiAl_{0\cdot9}$. The XRD spectrum for as cast $CoCrFeNiAl_{0\cdot9}$ is shown in Fig. 7*a*. The as cast



3 XRD pattern for as cast AlCoCrFeCu



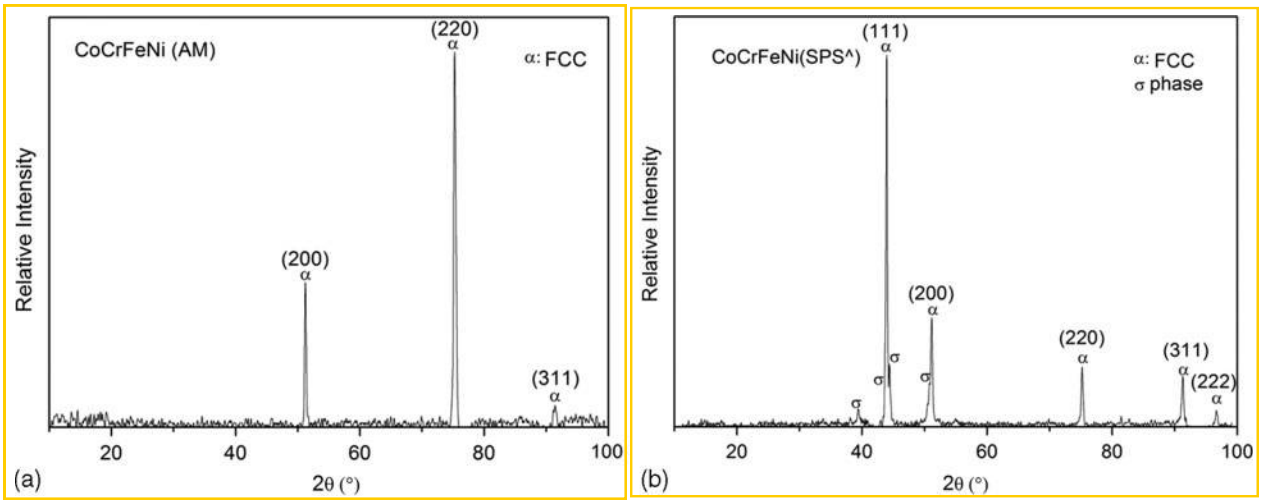
4 ***a*** **polarisation curves for as cast $CoCrFeNiAl_{0.9}$, CoCrFeNi and $Ti_{0.3}(CoCrFeNi)_{0.7}$ in 0.6 M NaCl at 25°C;** ***b*** **BSE image of $CoCrFeNiAl_{0.9}$ at low magnification;** ***c*** **BSE image of $CoCrFeNiAl_{0.9}$ at higher magnification;** ***d*** **BSE image of CoCrFeNi at low magnification;** ***e*** **BSE image of CoCrFeNi at higher magnification;** ***f*** **BSE image of $Ti_{0.3}(CoCrFeNi)_{0.7}$ at low magnification;** ***g*** **BSE image of $Ti_{0.3}(CoCrFeNi)_{0.7}$ at higher magnification**

Table 3 Electrochemical data associated with general corrosion and pitting corrosion of HEAs in 0.6 M NaCl at 25°C [CoCrFeNi (SPS^) was first hold at 500°C for 5 min and then sintered at 1000°C for another 5 min]

Sample	E_{corr}/mV_{SCE}	i_{corr}/μA cm^{-2}	E_{pit}/mV_{SCE})	Average $E_{pit}-E_{corr}$/mV
$CoCrFeNiAl_{0.9}$	−217±35	0·093±0·028	164±29	381
CoCrFeNi (SPS^)	−304±31	0·610±0·276	−8±59	260
$Ti_{0.3}(CoCrFeNi)_{0.7}$	−273±31	0·036±0·030	1040±13	1313

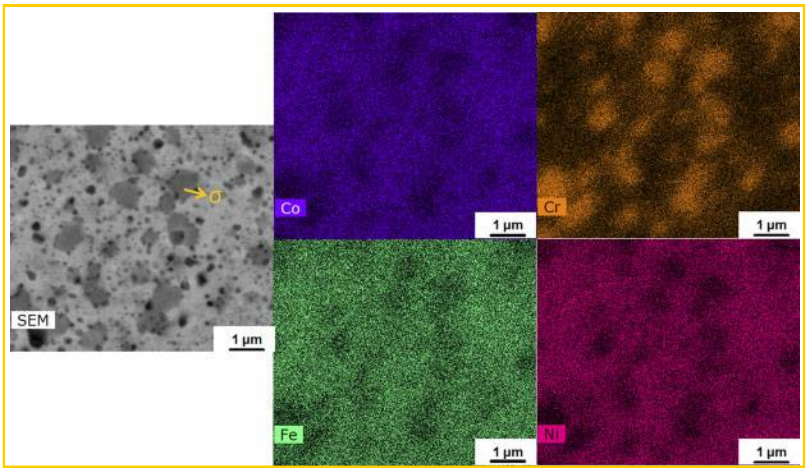
Table 4 EDS point analysis of phase 'A' of $Ti_{0.3}(CoCrFeNi)_{0.7}$ indicated in Fig. 4*g*

	Ti/at-%	Co/at-%	Cr/at-%	Fe/at-%	Ni/at-%
$Ti_{0.3}(CoCrFeNi)_{0.7}$	62·8	8·0	8·9	8·4	11·9

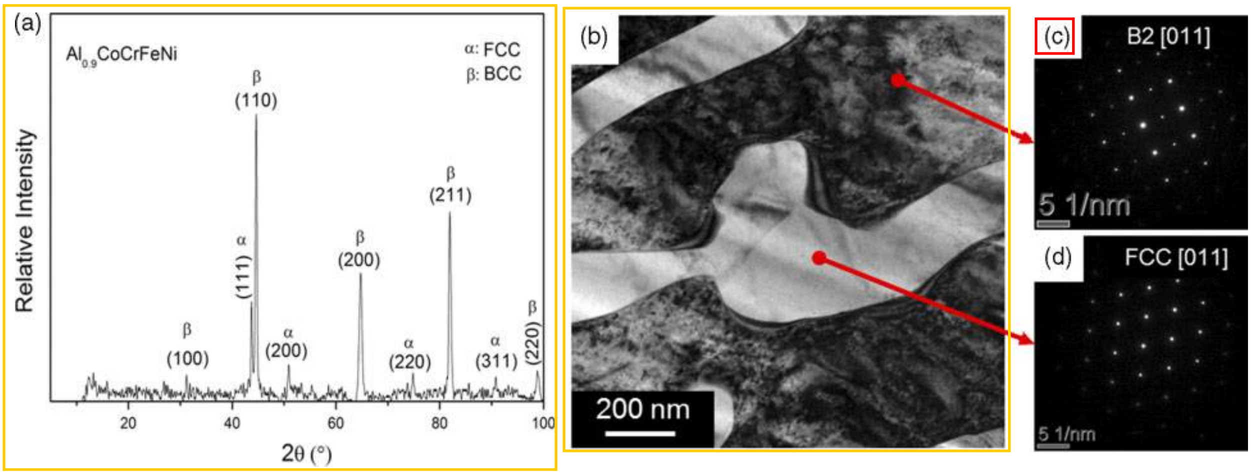


5 XRD pattern for ***a*** **as cast CoCrFeNi (AM) and** ***b*** **as cast CoCrFeNi (SPS^)**

$CoCrFeNiAl_{0.9}$ consists of a combination of fcc and bcc phase, not excluding that the latter may be in the form of the B2 phase (which is an ordered bcc phase). The bright field (BF) image of $CoCrFeNiAl_{0.9}$ is provided in Fig. 7*b*, and the select area diffraction pattern (SADP) of the dark contrast phase along the [011] zone axis is provided in Fig. 7*c* and is consistent with a B2 phase with a lattice parameter of 0·3001 nm. The SADP of the light contrast phase along the [011] zone axis in Fig. 7*d* is consistent with an fcc phase with



6 EDS mapping results for CoCrFeNi (SPS^) reveal that sigma phase is rich in Cr, while the matrix is Cr depleted



7 *a* XRD pattern for as cast $Al_{0.9}$CoCrFeNi; *b* TEM BF image of as cast $Al_{0.9}$CoCrFeNi; *c*, *d* SADP along [011] zone axis for 'dark' and 'light' phase respectively

a lattice parameter of 0·3704 nm. $CoCrFeNiAl_{0.9}$ exhibits a simple microstructure (fcc + B2), and there is no segregation of elements, like Cu in Cu containing HEAs or Cr in CoCrFeNi (SPS^), that could deteriorate the corrosion performance of the HEA. The simple microstructure and no elemental segregation could be associated with the relatively high corrosion potential and small corrosion current density of $CoCrFeNiAl_{0.9}$.

Avenues for future work

Future avenues of work in order to provide a more comprehensive determination of the corrosion of HEAs should pay particular attention to the following aspects

Multiscale characterisation

The microstructures of HEAs present features that span the atomic scale to the tens of micrometre scale (∼6 orders of magnitude). The area where the least characterisation exists to date is in the nanoscale, and therefore, HEAs should be examined by high resolution techniques such as atom probe tomography and high resolution TEM. Of the works to date that have studied nanostructure, useful insights have emerged, as one study[45] employs three-dimensional atom probe (3DAP), revealing the presence of fine Cu rich precipitates, Al–Ni rich plates and Cr–Fe rich features in cast AlCoCrCuFeNi HEA. Singh *et al.*[46] used 3DAP and TEM to study the ferromagnetic behaviour of AlCoCrCuFeNi HEAs, revealing the decomposition of CrFeCo rich regions into FeCo rich and Cr rich domains respectively.

Kinetics and nature of phase transformations

The majority of work to date that relates to HEAs (including the present work) have focused on 'as produced' alloys. This is in contrast to the very careful thermomechanical processing, which is applied to the study of common structural metals. In terms of simplicity, thermogravimetric analysis needs to be performed on HEAs to study the kinetics of, and characteristic temperature of, phase transformation. While phase diagrams for HEAs would be beneficial for understanding the kinetics and nature of phase transformations, existing CALPHAD (calculated phase diagram) databases are presently too primitive to allow for accurate HEA phase prediction. Indeed, however, as has been shown,[47] CALPHAD will be a very important tool to calculate the phase fields and fractions for HEAs, with some success reported for the AlCoCrFeNi and AlCoCrFeNi systems.

Surface film characterisation

The surface films formed upon different HEAs have been characterised using X-ray photoelectron spectroscopy (XPS) by several researchers.[17,20,31] Such studies, however, have focused on the surface film following dissolution/polarisation. In one instance, both $Al_{0\cdot5}CoCrCuFeNi$ and $Al_{0\cdot5}CoCrCuFeNiB_{0\cdot6}$ were analysed following anodic polarisation in a deaerated 1N H_2SO_4 solution. For these alloys, the predominant surface oxide was a Cr_2O_3 based film. A native Cr_2O_3 film before dissolution would indeed contribute to beneficial pitting resistance, as typically demonstrated by such HEAs.[20] The $CrFe_{1\cdot5}MnNi_{0\cdot5}$ HEA also forms a Cr_2O_3 based passive film on the surface after anodic polarisation in a 15% H_2SO_4 solution (as detected by XPS).[31] $Al_{0\cdot3}CrFe_{1\cdot5}MnNi_{0\cdot5}$ showed characteristic peaks corresponding to both Al_2O_3 and Cr_2O_3 (by XPS) after anodic treatment in a 15% H_2SO_4 solution.[31] It is not expected, however, that the Al_2O_3 can form as a native oxide (in conjunction with, or in addition to, Cr_2O_3). The Al_2O_3 is most likely formed due to preferential Al dissolution, then subsequent oxidation in the aqueous environment or air. Such a phenomenon of Al_2O_3 being detected in conjunction with Cr_2O_3 is common to Cr and Al containing superalloys (nominally Ni based superalloys) following high temperature oxidation.[48–51] In such instances, the oxidation occurs in air (at high temperatures) and preferential oxidation of Al occurs, somewhat akin to what occurs due to preferential Al dissolution in aqueous electrolytes. For example, in the case of AlCoCrFeNi coatings, after anodic polarisation in 3·5 wt-% NaCl, both Al_2O_3 and Cr_2O_3 peaks are observed in XPS spectra.[17] In any case, however, the XPS analysis of 'as polished' HEAs is particularly important to understand and is a particular aspect of HEAs that has essentially received no attention to date and must form the basis for future work. It has been observed herein that Ti improves the pitting resistance of the HEAs and also widens their passive window. The detection and quantification of Ti in passive films formed upon Ti based HEAs will also be important to study.

Long term exposure testing

In terms of studying the corrosion morphology, and the longer term damage accumulation (as a ground truth measurement), long term exposure testing is important. In terms of atmospheric corrosion resistance, because HEAs present passivity in the realm of stainless steels, atmospheric exposure will require very long term durations (i.e. years). In the laboratory setting, immersion in concentrated electrolytes is one avenue that is also beneficial. Kao *et al.*[22] and Hsu *et al.*[19] have reported mass loss/immersion tests on the $Al_xCoCrFeNi$ and $FeCoNiCrCu_x$ systems respectively. For $Al_xCoCrFeNi$ (where $x = 0$, 0·25, 0·5 and 1), immersion test were conducted in 0·5 M H_2SO_4 for 15 days, with mass loss revealing that the corrosion rate of $Al_{0\cdot5}CoCrFeNi$ ($\sim 1\ g\ m^{-2}\ h^{-1}$) and AlCoCrFeNi ($\sim 2\ g\ m^{-2}\ h^{-1}$) is significantly higher than that of CoCrFeNi ($0\cdot12\ g\ m^{-2}\ h^{-1}$) and $Al_{0\cdot25}CoCrFeNi$ ($0\cdot16\ g\ m^{-2}\ h^{-1}$). $FeCoNiCrCu_x$ alloys have been immersed in 3·5 wt-% NaCl at 25°C for 30 days, with the corrosion rate of $FeCoNiCrCu_x$ calculated from mass loss and i_{corr} obtained from polarisation tests revealing the same trend, in that the addition of Cu increases the corrosion rate.[19] It is, however, necessary to carry out longer term exposure testing in order to understand the general corrosion behaviour of HEAs.

Conclusions

1. A galvanic series in 0·6 M NaCl for select HEAs produced by either LM, AM or SPS has been presented. It is seen that the HEAs tested are significantly nobler than carbon steel and aluminium alloys, occupying potentials in the range of ferritic and austenitic stainless steels, with some HEAs nobler than austenitic stainless steels.
2. All the HEAs tested herein displayed 'spontaneous passivity' in 0·6 M NaCl. The pitting potentials, however, varied, with a range of passive windows ($E_{pit} - E_{corr}$) varying by as much as 1 V. The passivation tendency of HEAs was evident, in spite of possessing significantly heterogeneous microstructures.
3. Following from the point above, it should be noted that the microstructures evolved in the HEAs studied herein were nominally multiphase, with significant solute segregation. This aspect requires future work on the basis that characterisation is required to establish the kinetics of phase transformations, driven by partitioning, which is difficult to predict from existing thermodynamic calculations or conventional metallurgical systems.
4. In general, the addition of Cr to HEA systems improves their corrosion and pitting resistance. However, this benefit of Cr is severely decreased in HEAs processed by SPS (when compared to AM), as the Cr segregates to the Cr rich σ phase, making the Cr depleted matrix more prone to corrosion. The addition of Ti to the CrCoFeNi (AM) system was found to further increase the pitting corrosion resistance.
5. Variations in the electrochemistry and passive window of the AlCoCrCuFe system were attributed to disadvantageous Cu segregation to interdendritic regions.
6. $CoCrFeNiAl_{0\cdot9}$ displays a highly noble corrosion potential and low corrosion current density while possessing a simple structure of (fcc + B2) with no severe elemental segregation.

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